Data science portfolio

Andrei Chekunov

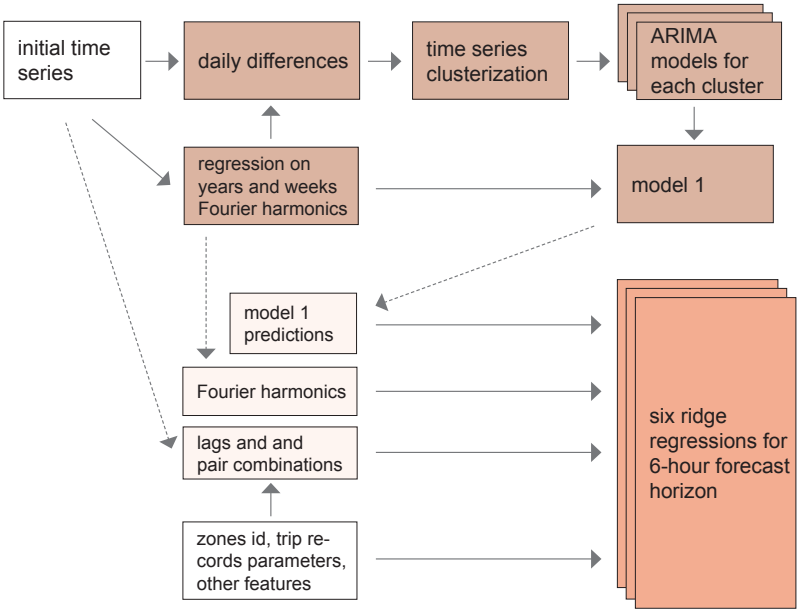
Sort-run prediction of yellow taxi trips

The final Project of ML and Data Analysis Specialisation form Yandex & Moscow Institute of Physics and Technology

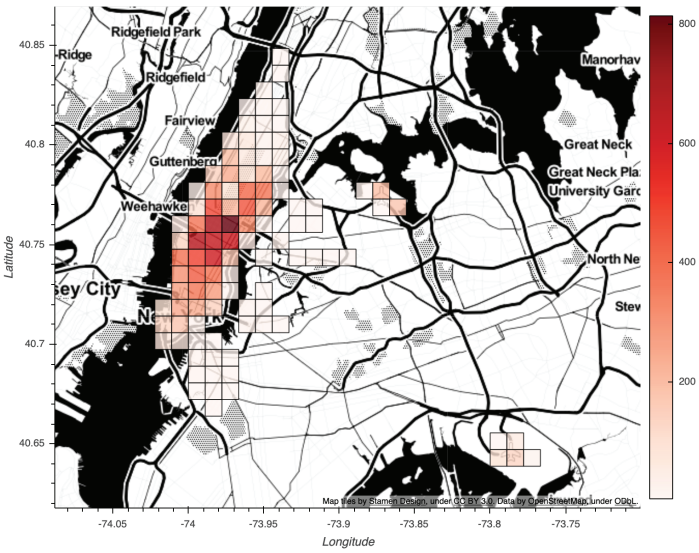
Problem: Multiple sort-run forecast of yellow taxi trips from high demand zones of New York.

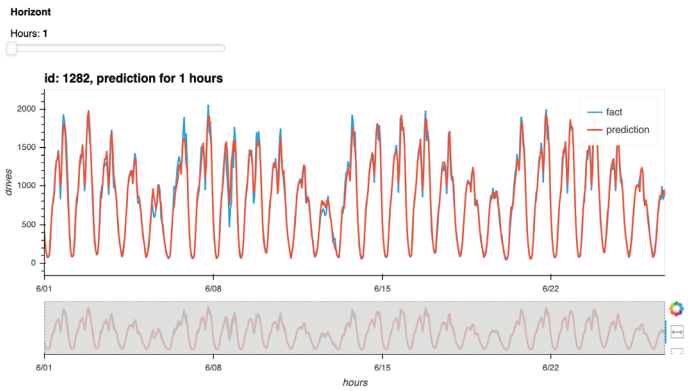
Data description: Yellow taxi trip records for more than 3 years: pick-up and drop-off dates/times, pick-up and drop-off locations, trip distances, rate types, etc. All trip coordinates were visualised on the New York map and checked on plausibility, also zones of high demand were detected.
https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page

Two level Model: On the first step was used an ensemble of regression on Fourier harmonics for year and week components and few typical ARIMA models for daily rests. Tentatively all daily time series were clustered and for the most typical representative of each cluster was tuned an individual ARIMA. On the second step was used a ridge regression on prediction of Model 1, Fourier harmonics; Zones id, trip records parameters, lags and and different pair combinations; other features, such as Airport dummy, number of trips in the zone and out the neighbours zones etc.



Results representation: An interactive widget with possibility for selection of each zone forecast for each hour of six-hours horizon.





Predict Future Sales

The final project of “How to Win a Data Science Competition” Course in the Advanced ML Specialisation from HSE

Problem: Next month prediction of total sales (item / store pairs) of the largest Russian software firm - 1C Company.
https://www.kaggle.com/c/competitive-data-science-predict-future-sales/overview

Result: 121th place out of about 11 000 teams on Kaggle, 0.856 rmse
(naïve benchmark: 1.16, the best score: 0.75)

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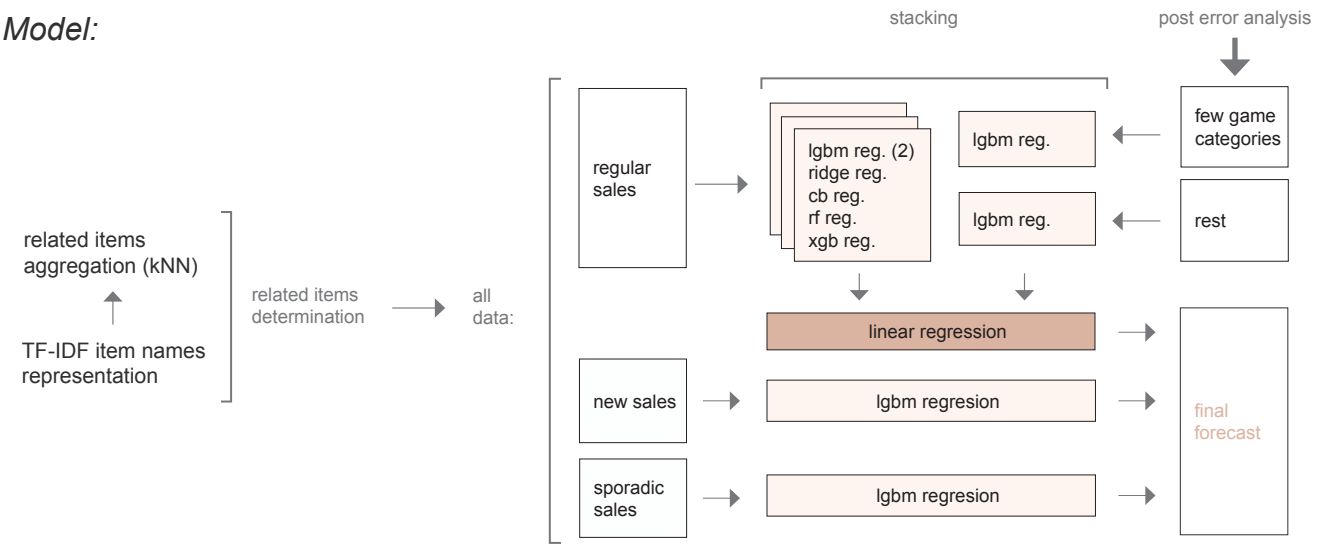
Data: Three year daily transactions: about 60 shops / 21 000 items; items description: text, category; shops description: text

Main steps: Data analysis / segmentation, feature engineering, modelling / stacking

Feature engineering:

Source / Target	Regular sales:	New sales:
Sales and Calendar based features:	Lags and EMA on adjusted monthly sales with respect of the number of working days/weekends and holidays Days on sale for each item/shop; days from the last sale	Months/days on sale of each item
Geography based features:	Lags on sales of other shops in the same city Differences between the shop and weighted mean of other shops sales in the same city Lags on sales of related items (shop, overall), shop/category sales structure	
Aggregations and statistics:	Lags on sale frequencies Different kinds of aggregations and differences: item, shop, category, grouped categories; cumulative sales	Shop/category priority indicators Average first sale for pairs category/shop
Others:	Months, kinds of shop, shop age, cities, price trends, etc.	
Features extracted from item description:	Attributes: platforms, language, digital, etc. Related items: lags, cum sum (shop, overall)	First sales of related items (shop, overall)

Model:



Intruder Detection through Webpage Session Tracking

The final Project of ML and Data Analysis Specialisation form Yandex & Moscow Institute of Physics and Technology

Problem: Identification of a specific user by tracking logs of attended Web pages.
https://www.kaggle.com/c/catch-me-if-you-can-intruder-detection-through-webpage-session-tracking2/

Result: 10th place out of about 4700 teams on Kaggle; 0.972 roc-auc.
(benchmark: simple BOW on sessions: 0.916, the best score: 0.985)

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Data: Logs (site name, time of visit) of about, 1500 users for 16 months.

Solution: https://github.com/olip78/Catch-Me-If-You-Can

Other projects

The final project of “Bayesian Methods for Machine Learning” Course in the Advanced ML Specialisation from HSE

Facial Composites. Bayesian optimization of a similarity function between a person’s face in witness’s memory and a face reconstructed from the current point of latent space of a pertained Variational Autoencoder,

The final project of “Introduction to Deep Learning” Course in the Advanced ML Specialisation from HSE

Image Captioning. An ensemble of vision deep learning CNN and language generating RNN

The final project of “Natural Language Processing” Course in the Advanced ML Specialisation from HSE

StackOverflow assistant bot, a conversational assisting with search on StackOverflow

Final Projects of ML and Data Analysis Specialisation form Yandex & Moscow Institute of Physics and Technology

Customer churn prediction on blind data of a telecommunication company

Certificates / Courses

Advanced Machine Learning Specialisation, Higher School of Economics, Moscow
https://www.coursera.org/specializations/aml

5 / 7 Courses:

Introduction to Deep Learning
How to Win a Data Science Competition: Learn from Top Kagglers (with Honors)
Bayesian Methods for Machine Learning (with Honors)
Addressing Large Hadron Collider Challenges by Machine Learning
Natural Language Processing

Machine Learning and Data Analysis Specialisation, Moscow Institute of Physics and Technology & Yandex
https://www.coursera.org/specializations/machine-learning-data-analysis

6 Course Specialisation:

Math and Python,
Supervised Learning,
Unsupervised Learning and Data Visualisation,
Statistics for Data Science,
Practical Data Analysis,
Final Project

Certificate

Industrial Machine Learning Specialisation, Higher School of Economics
https://www.coursera.org/specializations/aml

2 / 3 Courses:

Big Data Processing (Spark, Hadoop, Map Reduce, Mongo DB, SQL, Bash)
Machine Learning on Big Data (ETL, Vowpal Wabbit, Spark, H2O, Feature Hashing, Scaling Computations, Distributed Learning, Transfer Learning etc.)

Other Courses

Data Visualization and Communication with T,ableau, Duke University
Databases and SQL for Data Science with Python, IBM